Tareas a Realizar Tarea 1:

Configuración Global y Reenviadores

Edita el fichero de opciones globales de BIND9 (/etc/bind/named.conf.options) para cumplir con las siguientes especificaciones:

sudo apt update

sudo apt install bind9 bind9utils

```
ri-server@sri-server:/etc/bind$ ls
amed.conf  named.conf.local  named.conf.options  named.conf.root-hints  rndc.key
ri-server@sri-server:/etc/bind$
```

/etc/bind/named.conf.options

1. Configura reenviadores (forwarders) a las IPs 8.8.8.8 y 1.1.1.1 .

2. Permite las consultas DNS (allow-query) exclusivamente a la red local 192.168.100.0/24 y a la interfaz local (localhost).

```
options {
        directory "/var/cache/bind";
        forwarders {
                8.8.8.8;
                1.1.1.1;
        };
        allow-query {
                localhost;
                192.168.100.0/24;
        };
};
```

sudo named-checkconf - revisar si todo bien

Tarea 2: Declaración y Fichero de Zona Directa

1. Declara la zona directa techcorp.lan en el fichero /etc/bind/named.conf.local

sudo nano /etc/bind/named.conf.local

```
  GNU nano 8.7.1
//
// Do any local configuration here
//
zone "techcorp.lan" {
        type master;
        file "/etc/bind/db.techcorp.lan";
};
```

sudo named-checkconf

2. Crea el fichero de zona correspondiente (/etc/bind/db.techcorp.lan) e incluye los siguientes registros:

sudo nano /etc/bind/db.techcorp.lan

3. SOA y NS: Servidor de nombres principal ns1.techcorp.lan . Email de contacto: admin.techcorp.lan .

sudo nano /etc/bind/db.techcorp.lan

4. Registros A: ns1 -> 192.168.100.10 gw -> 192.168.100.1 web -> 192.168.100.20 mail -> 192.168.100.30

5. Registro CNAME: www que apunte al equipo

6. Registro MX: web.techcorp.lan . techcorp.lan con prioridad 10 apuntando a mail.techcorp.lan .

```
$TTL 604800

@       IN      SOA     ns1.techcorp.lan.       admin.techcorp.lan (
                        2026101001      ; Serial
                        604800          ; Refresh
                        86400           ; Retre
                        2419200         ; Expire
                        604800 )        ; Nevative Cache TTL
;

@       IN      NS      ns1.techcorp.lan.
@       IN      MX      10 mail.techcorp.lan.

ns1     IN      A       192.168.100.10
gw      IN      A       192.168.100.1
web     IN      A       192.168.100.20
mail    IN      A       192.168.100.30

www     IN      CNAME   web.techcorp.lan.
```

```
sri-server@sri-server:~$ sudo named-checkzone techcorp.lan /etc/bind/db.techcorp.lan
zone techcorp.lan/IN: loaded serial 2026101001
OK
sri-server@sri-server:~$
```

Tarea 3: Declaración y Fichero de Zona Inversa

1. Declara la zona de resolución inversa correspondiente a la red 192.168.100.0/24 (100.168.192.in-addr.arpa) en /etc/bind/named.conf.local .

sudo nano /etc/bind/named.conf.local

```
//
// Do any local configuration here
//
zone "techcorp.lan" {
        type master;
        file "/etc/bind/db.techcorp.lan";
};
zone "100.168.192.in-addr.arpa" {
        type master;
        file "/etc/bind/db.192.168.100";
};
```

sudo named-checkconf

2. Crea el fichero de zona inversa (/etc/bind/db.192.168.100) e incluye los siguientes registros:

3. SOA y NS: Mismos parámetros que la zona directa.

4. Registros PTR: 10 -> ns1.techcorp.lan. 1 -> gw.techcorp.lan. 20 -> web.techcorp.lan. 30 -> mail.techcorp.lan.

```
$TTL 604800
@       IN      SOA     ns1.techorp.lan.        admin.techcorp.lan. (
                        2026101001      ;       Serial
                        604800          ;       Refresh
                        86400           ;       Retry
                        2419200         ;       Expire
                        604800  )       ;       Negative Cache TTL
;
@       IN      NS      ns1.techcorp.lan.

10      IN      PTR     ns1.techcorp.lan.
1       IN      PTR     gw.techcorp.lan.
20      IN      PTR     web.techcorp.lan.
30      IN      PTR     mail.techcorp.lan.
```

```
sri-server@sri-server:/etc/bind$ sudo named-checkzone 100.168.192.in-addr.apra /etc/bind/db.192.168.100
zone 100.168.192.in-addr.apra/IN: loaded serial 2026101001
OK
sri-server@sri-server:/etc/bind$
```

Tarea 4: Comprobación y Verificación

1. Verifica la sintaxis de las zonas y de los ficheros de configuración mediante las herramientas nativas de BIND9

sudo nano /etc/bind/named.conf.local

sudo named-checkconf

```
//
// Do any local configuration here
//
zone "techcorp.lan" {
        type master;
        file "/etc/bind/db.techcorp.lan";
};
zone "100.168.192.in-addr.arpa" {
        type master;
        file "/etc/bind/db.192.168.100";
};
```

```
sri-server@sri-server:/etc/bind$ sudo named-checkconf
sri-server@sri-server:/etc/bind$
```

sudo named-checkzone 100.168.192.in-addr.arpa /etc/bind/db.192.168.100

```
sri-server@sri-server:/etc/bind$ sudo named-checkzone 100.168.192.in-addr.apra /etc/bind/db.192.168.100
zone 100.168.192.in-addr.apra/IN: loaded serial 2026101001
OK
```

sudo named-checkzone techcorp.lan /etc/bind/db.techcorp.lan

```
sri-server@sri-server:/etc/bind$ sudo named-checkzone techcorp.lan /etc/bind/db.techcorp.lan
zone techcorp.lan/IN: loaded serial 2026101001
OK
sri-server@sri-server:/etc/bind$
```

2. Reinicia el servicio named (o bind9).

sudo systemctl restart bind9

sudo systemctl status bind9

```
sri-server@sri-server:/etc/bind$ sudo named-checkconf
sri-server@sri-server:/etc/bind$ sudo systecmctl restart bind9
sudo: 'systecmctl': command not found
sri-server@sri-server:/etc/bind$ cd
sri-server@sri-server:~$ sudo systemctl restart bind9
sri-server@sri-server:~$ sudo systemctl status bind9
● named.service - BIND Domain Name Server
     Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-11-03 20:05:53 UTC; 20s ago
 Invocation: 64858b2831eb448fa37f19e08fb6808f
       Docs: man:named(8)
   Main PID: 25993 (named)
     Status: "running"
      Tasks: 6 (limit: 1833)
     Memory: 6.7M (peak: 6.7M)
        CPU: 203ms
     CGroup: /system.slice/named.service
             └─25993 /usr/sbin/named -f -u bind

nov 03 20:05:53 sri-server named[25993]: network unreachable resolving './NS/IN': 2001:500:12::d0d#53
nov 03 20:05:53 sri-server named[25993]: network unreachable resolving './NS/IN': 2001:7fe::53#53
nov 03 20:05:53 sri-server named[25993]: network unreachable resolving './NS/IN': 2001:7fd::1#53
nov 03 20:05:53 sri-server named[25993]: network unreachable resolving './NS/IN': 2001:500:2::c#53
nov 03 20:05:53 sri-server named[25993]: network unreachable resolving './NS/IN': 2001:500:9f::42#53
nov 03 20:05:53 sri-server named[25993]: network unreachable resolving './NS/IN': 2001:500:1::53#53
nov 03 20:05:53 sri-server named[25993]: managed-keys-zone: Key 20326 for zone . is now trusted (acceptance
nov 03 20:05:53 sri-server named[25993]: managed-keys-zone: Key 38696 for zone . is now trusted (acceptance
nov 03 20:06:09 sri-server named[25993]: zone 100.168.192.in-addr.arpa/IN: notify to 192.168.100.10#53 faile
nov 03 20:06:09 sri-server named[25993]: zone 100.168.192.in-addr.arpa/IN: sending notify to 192.168.100.10#
sri-server@sri-server:~$
```

Configuración de la red interna para asignar al servidor DNS la IP 192.168.100.10.

```
  GNU nano 8.7.1
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      dhcp4: true
      dhcp6: true
      match:
        macaddress: 08:00:27:66:0a:e2
      set-name: enp0s3

    enp0s8:
      dhcp4: false
      addresses:
          - 192.168.100.10/24
  version: 2
```

3. Desde un equipo cliente o desde el propio servidor, ejecuta las siguientes consultas con dig o nslookup guardando las capturas o salidas de texto en el informe de entrega:

dig @192.168.100.10 techcorp.lan MX +short

```
sri-server@sri-server:~$ dig @192.168.100.10 techcorp.lan MX +short
10 mail.techcorp.lan.
```

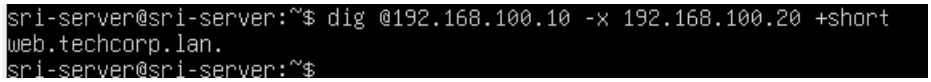
4. Consulta de dirección A para www.techcorp.lan .

dig @192.168.100.10 www.techcorp.lan +short

```
sri-server@sri-server:~$ dig @192.168.100.10 www.techcorp.lan +short
web.techcorp.lan.
192.168.100.20
sri-server@sri-server:~$
```

5. Consulta inversa PTR para la IP 192.168.100.20 .

dig @192.168.100.10 -x 192.168.100.20 +short



Enlace al vídeo: sriud2actividad_2.mp4